# Ready72 – AI-Powered Personalized Earthquake Preparedness Assistant

## Problem and Solution

Generic earthquake preparedness checklists often provide the same advice to everyone, although household needs differ according to age groups, mobility limitations, pets, housing conditions, existing supplies, and location context.

Ready72 addresses this problem by using generative AI to create a personalized earthquake preparedness plan. Users enter basic household information such as city, district, building type, floor, household members, mobility needs, pets, and existing emergency supplies.

The system generates:

- a household preparedness summary
- priority actions
- role assignments
- a personalized emergency kit
- a first 72 hours action plan

The project primarily contributes to SDG 11 – Sustainable Cities and Communities, especially Target 11.5, which focuses on reducing the impact of disasters on people and communities.

## Target User and Success Criteria

Ready72 is designed for non-expert households in Türkiye that want a plan matching their own home and family rather than a generic checklist. Besides SDG 11.5, it supports SDG 13 through disaster resilience at household level.

We judge whether the tool works by (a) automatic checks on every plan (structure, a task for every member, no prohibited content); (b) manual review on four contrasting test profiles (young child, elderly member with limited mobility, pet owner, person living alone), checking that each section uses household-specific details; and (c) comparing three prompt variants: A (baseline), B (two-stage chain) and C (chain with few-shot examples).

## Generative AI and Prompt Design

The current prototype uses Google Gemini 2.5 Flash through OpenRouter as the default model configuration.

Ready72 uses a default two-stage prompting workflow. First, the model analyzes household characteristics, assistance needs, missing supplies, and preparedness priorities. Then, it generates the final personalized plan.

Few-shot examples and structured JSON outputs are used to improve consistency.

The application also performs rule-based validation. It checks required sections, item counts, relevant household role coverage, selected prohibited expressions, and the presence of five time stages.

If validation problems occur, the system allows one repair attempt for the analysis stage and one repair attempt for the plan stage. However, validation does not guarantee that every recommendation is fully correct, personalized, or safe. Remaining issues may still be stored as internal warnings.

**Prompt techniques and why we use them:**

- **Zero-shot:** Stage 1 (analysis) and baseline Prompt A use only a clear task and schema; the baseline shows what a naive prompt produces.
- **Few-shot:** Stage 2 (plan) includes three example households showing structure, tone and the "Item — household-specific reason" format, with a "do not copy" rule and a code check against copied sentences. Stage 1 has no examples, to avoid anchoring.
- **Step-by-step reasoning:** the two-stage chain separates analysis from writing; each task prompt also lists numbered steps and a PASS/FAIL self-check.
- **Negative prompting and injection defense:** DO/DON'T rules (form, content, behavior); priority order system > task > data; user data is wrapped in tags and treated only as data.

**Key prompts** (Turkish, short excerpts; full prompts are in prompts_tr.py):

- **System prompt:** *"Sen Ready72'sin: … sakin ve net bir planlama asistanısın. Yetkili kurum veya afet uzmanı değilsin."* (You are a calm planning assistant, not an authority.) Rules forbid invented assembly areas, routes and phone numbers, building judgments, predictions, guarantees and drug doses.
- **Stage 1:** *"Her kişi etiketine tek bir sorumluluk ata. … Bilinmeyen/doğrulanması gereken noktaları işaretle."* (Give each person label one duty; flag unknowns.)

- **Stage 2:** *"<hane_analizi> tek doğru kaynaktır; … analizde olmayan kişi veya görev uydurma."* (The analysis is the only source of truth; do not invent people or tasks.) Few-shot rule: *"Cümleleri birebir kopyalama."* (Do not copy sentences.)

### Challenges and Solutions

- **Generic, repetitive plans:** a single prompt tended to give similar advice to every household and overlapping phases. We split analysis from writing, required household-specific details in every section and reject near-identical phases.
- **Invented or inconsistent labels:** person and phase labels are computed in code, and every member must receive a task.
- **Risky or misplaced advice:** gas-valve advice may appear only in the 1–15 minute phase and only conditionally; risky phrases such as "completely safe" or "use the elevator" are blocked by a list.
- **Copying and irrelevant content:** validators detect copied example sentences and pet mentions for households without pets, then one repair attempt runs. Low temperature, a fixed seed and caching keep results repeatable.

### Safety and Ethics

Because earthquake preparedness is a high-impact domain, Ready72 applies explicit safety restrictions.

The AI is instructed not to generate:

- official assembly areas
- evacuation routes
- emergency phone numbers
- building safety assessments
- unsupported local risk claims
- earthquake predictions
- guarantees of safety

When official information is needed, users are directed to trusted sources such as AFAD, e-Devlet, or local authorities.

The model itself does not generate emergency phone numbers; 112 is added by the application as fixed text.

**Vulnerable groups:** children are handled by age group (0–2 are carried, 3–5 stay with an adult, 6–12 receive simple tasks, 13–17 assist adults); evacuation is planned around the least mobile member; medication advice is limited to keeping a written list and a spare supply and asking a doctor, with no doses.

**Other risks:** to limit bias, the plan is built only from the entered profile (no assumed typical family, car or safe building), and an unknown building type leads to a referral to official inspection. To limit over-trust, users are told to verify with official sources, since validation checks structure and selected rules, not factual correctness.

Ready72 should also clearly display:

Ready72 supports earthquake preparedness planning and does not replace official emergency instructions or authorities.

### Limitations and Future Development

The current prototype does not yet retrieve verified local hazard data, official assembly areas, evacuation routes, or real-time alerts.

Future versions could integrate verified local risk data, official assembly areas, maps and routes, AFAD or municipality data, accessibility-aware routing, and emergency simulations.

### Conclusion

Ready72 shows how generative AI can move beyond one-size-fits-all preparedness checklists by combining household information with structured prompting, validation, and safety rules.

*Preparedness should not be one-size-fits-all.*